K.R. MANGALAM UNIVERSITY

Database Management Lab File
(ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

Aryan (2401010005)

B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

MS. Mansi Kajal

School of Engineering & Technology
K. R. MANGALAM UNIVERSITY
Sohna, Haryana 122103, India

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Experiment: 1

1.Consider following databases and draw ER diagram and convert entitiesand relationships to relation table for a given scenario:

Aim: To analyze the requirements for a College Database, design an Entity-Relationship (ER) Diagram representing Students, Sections, Subjects, and Marks, and convert the conceptual model into a Relational Database Schema.

COLLEGE DATABASE:

STUDENT (USN, SName, Address, Phone, Gender)

SEMSEC (SSID, Sem, Sec)

CLASS (USN, SSID)

SUBJECT (Subcode, Title, Sem, Credits)

IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

PROCEDURE:

Step 1: Create the Strong Entities (Rectangles) Draw four rectangles to represent the main tables in your database:

1. **STUDENT**
2. **SEMSEC** (Semester & Section)
3. **SUBJECT**
4. **IAMARKS** (Internal Assessment Marks)

Step 2: Add Attributes (Ovals) Draw ovals connected to their respective entities. Make sure to identify Primary Keys and special attributes.

- For STUDENT:
  - Draw ovals: SNAME, ADDRESS, PHONE, GENDER.
  - Draw oval USN and **underline** the text (Primary Key).
- For SEMSEC:
  - Draw ovals: SEM, SEC.
  - Draw oval SSID and **underline** the text (Primary Key).
- For SUBJECT:
  - Draw ovals: Title, SEM, CREDITS.
  - Draw oval SUBCODE and **underline** the text (Primary Key).

- For IAMARKS:
    - Draw ovals: Test1, Test2, Test3.
    - Draw a Dashed/Dotted Oval for FinalIA. (This indicates it is a Derived Attribute, calculated from the other test scores).

Step 3: Create Relationships (Diamonds) Draw four diamond shapes to connect the entities.

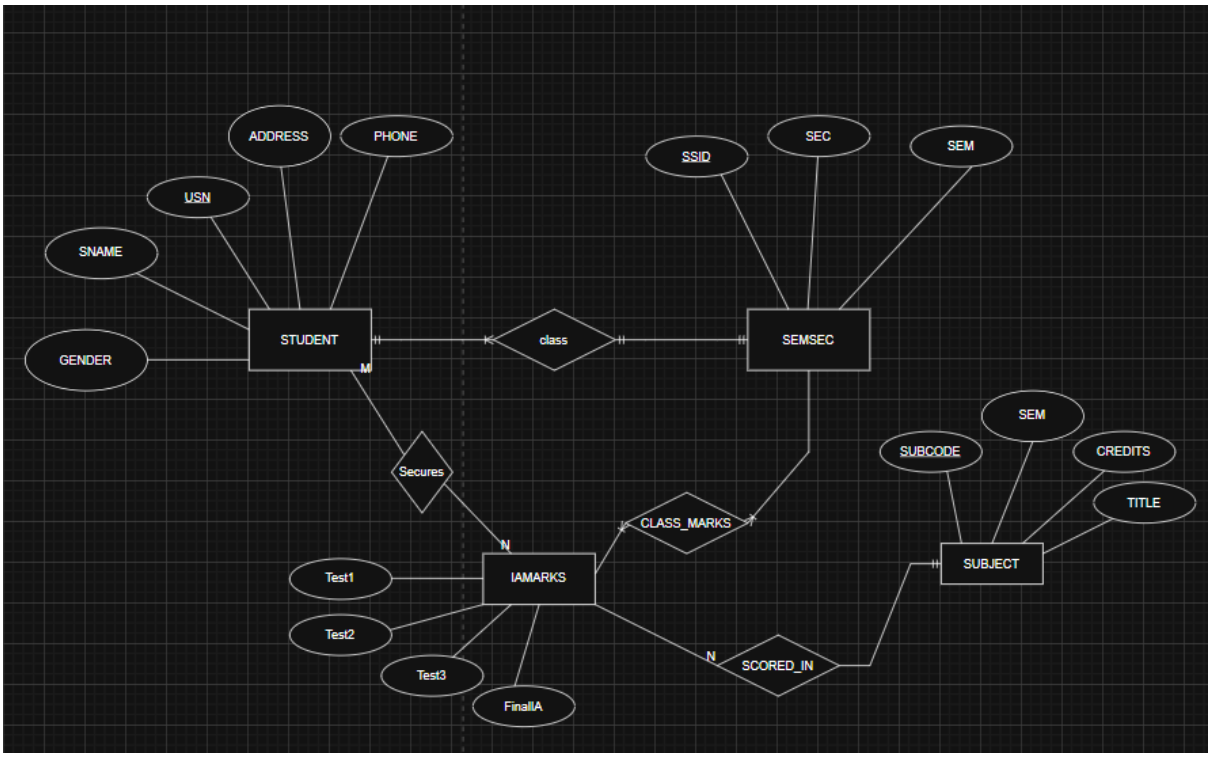
1. Draw "CLASS" Diamond:
    - Place it between STUDENT and SEMSEC.
    - Meaning: Represents which class/section a student belongs to.
2. Draw "Secures" Diamond:
    - Place it between STUDENT and IAMARKS.
    - Meaning: Represents the marks secured by a specific student.
3. Draw "CLASS_MARKS" Diamond:
    - Place it between SEMSEC and IAMARKS.
    - Meaning: Links the marks to a specific section (to identify which class the marks belong to).
4. Draw "SCORED_IN" Diamond:
    - Place it between SUBJECT and IAMARKS.
    - Meaning: Identifies which subject the marks are for.

Step 4: Connect Lines & Define Cardinality (Mapping) Draw lines connecting the diamonds to the rectangles and label the cardinality (1, M, N) as shown in your image.

- For CLASS (Student - SemSec):
    - Draw a single line from STUDENT (Label 1).
    - Draw a double line (or crow's foot) from SEMSEC (Label N or (1, n)).
    - *Logic:* One Student is in One Section; One Section has Many Students.
- For Secures (Student - IAMarks):
    - Connect STUDENT to IAMARKS.
    - Label as M: N (Many-to-Many).
- For SCORED_IN (Subject - IAMarks):
    - Connect SUBJECT to IAMARKS.
    - Label as 1: N (One Subject has Many Marks entries).

- For CLASS_MARKS (SemSec - IAMarks):
    - Connect SEMSEC to IAMARKS.
    - Label as 1: N (One Section has Many Marks entries).

Step 5: Final Review Check that FinalIA is dashed (derived), all Primary Keys (USN, SSID, SUBCODE) are underlined, and all 4 entities are connected through the central IAMARKS relationships



EXPERIMENT: 2

2. Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

AIM: To analyze the requirements for a Company Database, design an Entity-Relationship (ER) Diagram representing Employees, Departments, Projects, and their relationships (Works On, Manages, Supervision), and map the ER diagram into a normalized Relational Database Schema.

PROCEDURE:

Step 1: Draw the 3 Main Entities (Rectangles)

Draw three rectangular boxes in the following positions:

1. DEPARTMENT: Place it in the top-middle.
2. Employee: Place it on the right side.
3. PROJECT: Place it in the bottom-left.

Step 2: Add Attributes (Ovals)

Draw ovals connected to their respective rectangles. Crucially, include the Foreign Keys as shown in your image:

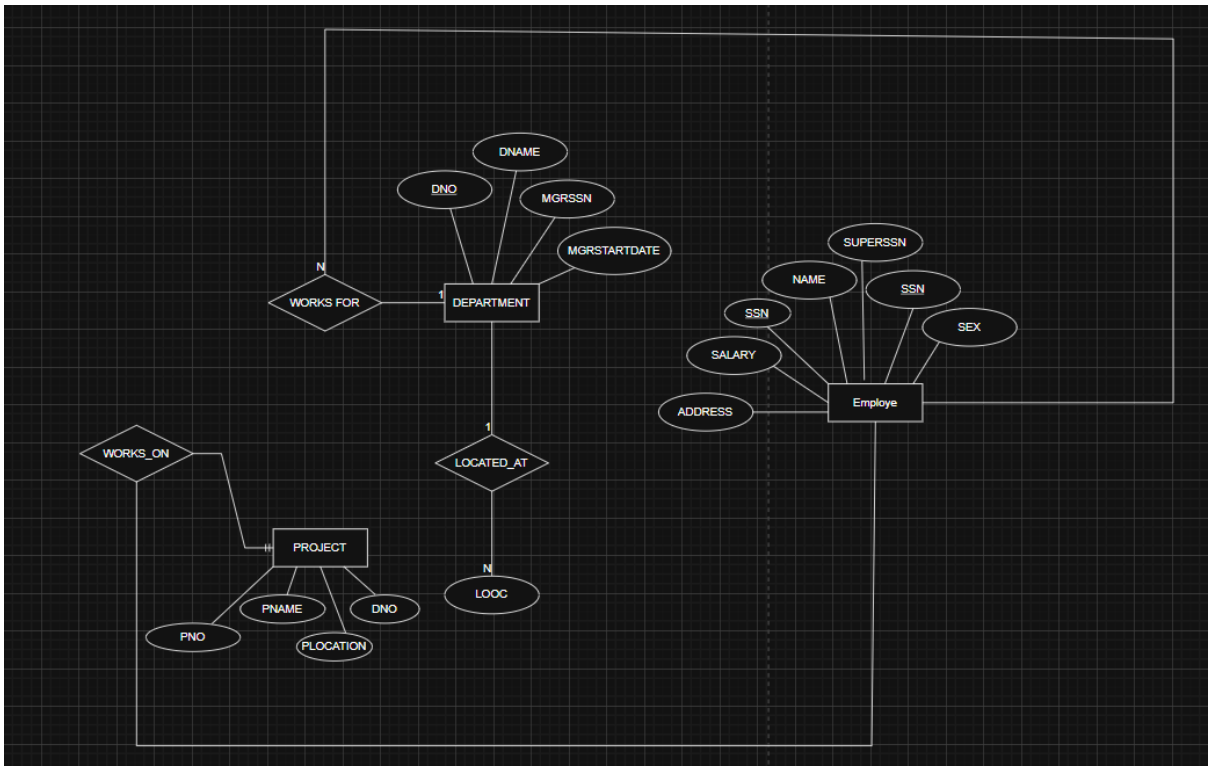
- For DEPARTMENT:
  - Add ovals: DNAME, MGRSSN, MGRSTARTDATE.
  - Add oval DNO and underline it (Primary Key).
- For Employee:
  - Add ovals: NAME, ADDRESS, SALARY, SEX.
  - Add oval SSN and underline it (Primary Key).
  - Add Foreign Keys: Add ovals for SUPERSSN and DNO attached to this entity.
- For PROJECT:
  - Add ovals: PNAME, PLOCATION.
  - Add oval PNO and underline it (Primary Key).
  - Add Foreign Key: Add an oval for DNO attached to this entity.
- For Locations:
  - Draw a separate oval labeled LOOC below the Department table.

Step 3: Draw Relationships (Diamonds)

Draw three diamond shapes to represent the connections:

1. WORKS_FOR: Place this between DEPARTMENT and Employee.
2. LOCATED_AT: Place this between DEPARTMENT and the LOOC oval.

3. WORKS_ON: Place this to the left of the PROJECT table.

Step 4: Connect Lines & Define Cardinality

1. Connect WORKS_FOR:
   - Draw a line from DEPARTMENT to the diamond. Label it 1 (or N depending on your specific image logic, usually Many Employees work for 1 Dept). *In your image, N is near Employee and 1 is near Dept.*
   - Draw a line from Employee to the diamond**.**
2. Connect LOCATED_AT:
   - Draw a line from DEPARTMENT down to the diamond. Label it 1.
   - Draw a line from the diamond down to the LOOC oval. Label it N.
3. Connect WORKS_ON:
   - Draw a line from PROJECT to the diamond.
   - Draw a long line from the other side of the diamond that goes Up, turns Right (across the top of the page), turns Down, and connects to the Employee table.
   - This represents the Many-to-Many relationship between Employees and Projects.